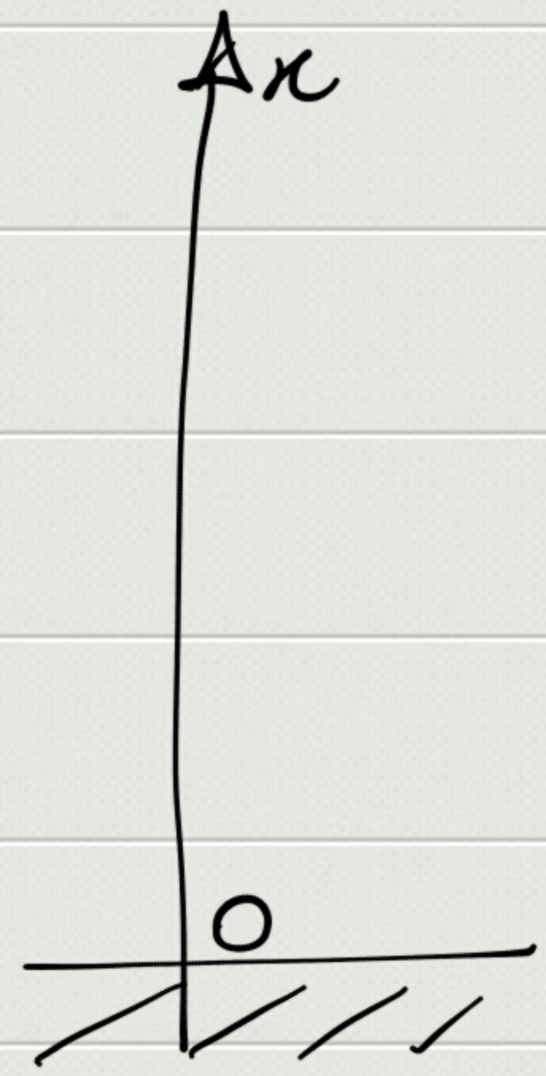
$$g = 9.81 \ m/s^2$$



$$a = -g = -9.81 \ m/s^2$$

$$\boxed{a = \text{cost}}$$

$$x(t) = x_0 + v_0 (t - t_0) + \frac{1}{2} a (t - t_0)^2$$

$$v(t) = v_0 + a (t - t_0)$$

$$v^2(x) = v_0^2 + 2a (x - x_0)$$

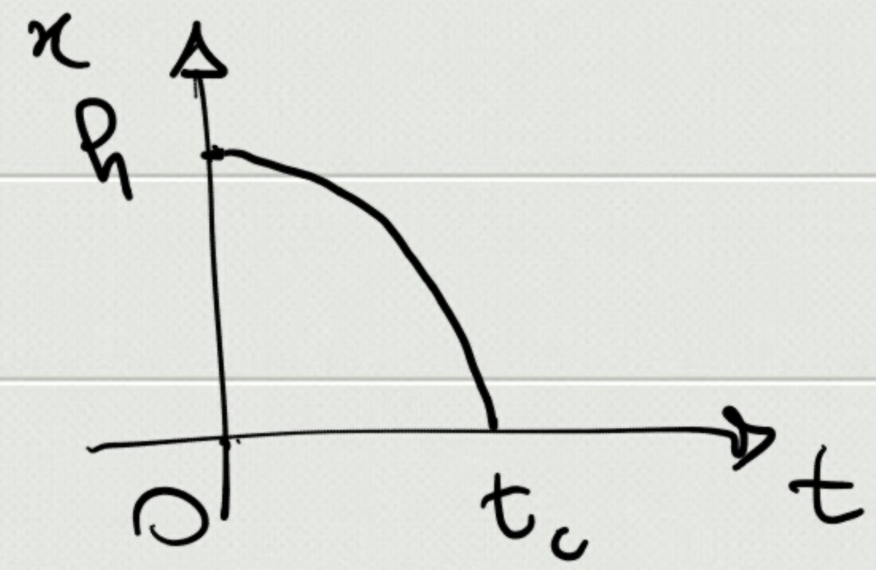
$$t_0 = 0 \qquad a = -g$$

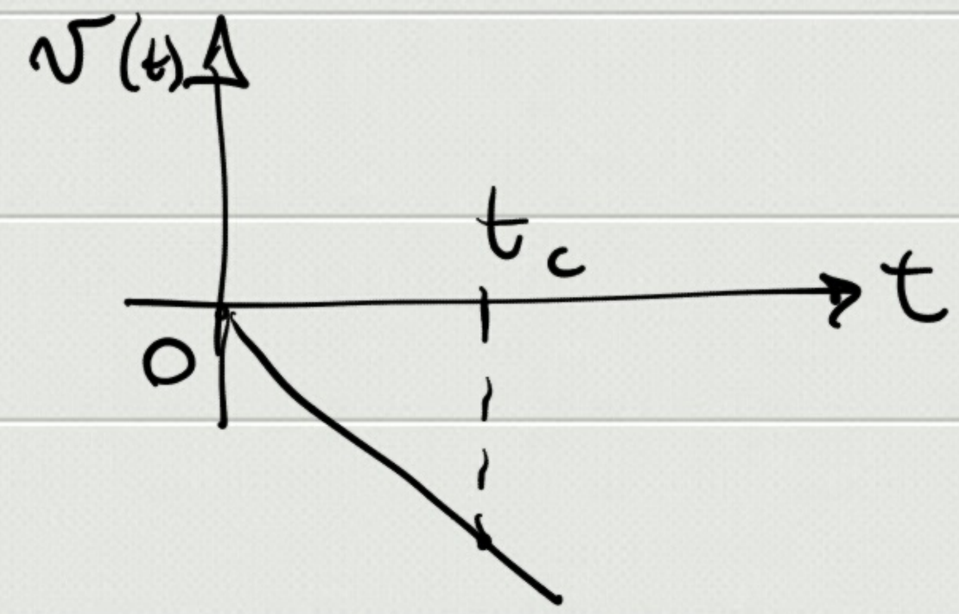
$$\boxed{x(t) = x_0 + v_0 t - \frac{1}{2} g t^2}$$

$$\boxed{v(t) = v_0 - g t} \qquad \boxed{v^2(x) = v_0^2 - 2g (x - x_0)}$$

$x_0 = h > 0 \qquad v_0 = 0$

$$\boxed{\begin{aligned} x(t) &= h - \frac{1}{2} g t^2 \\ v(t) &= -g t \\ v^2(x) &= -2g(x-h) \end{aligned}}$$





$$t(x) = \sqrt{\frac{2(h-x)}{g}} \qquad t_c = t(x=0) = \sqrt{\frac{2h}{g}}$$

$$v_c = v(t_c) = -g\sqrt{\frac{2h}{g}} = -\sqrt{2gh}$$

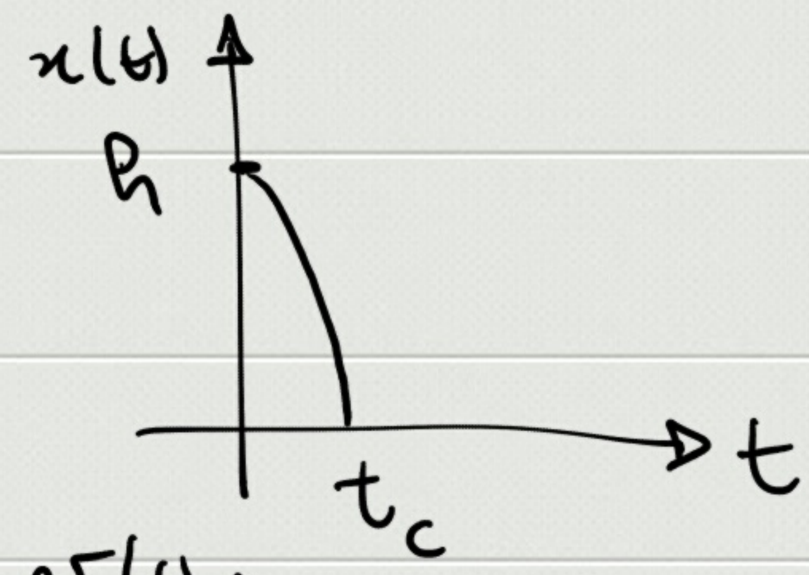
$$v(x) = -\sqrt{2g(h-x)} \Rightarrow v_c = v(x=0) = -\sqrt{2gh}$$

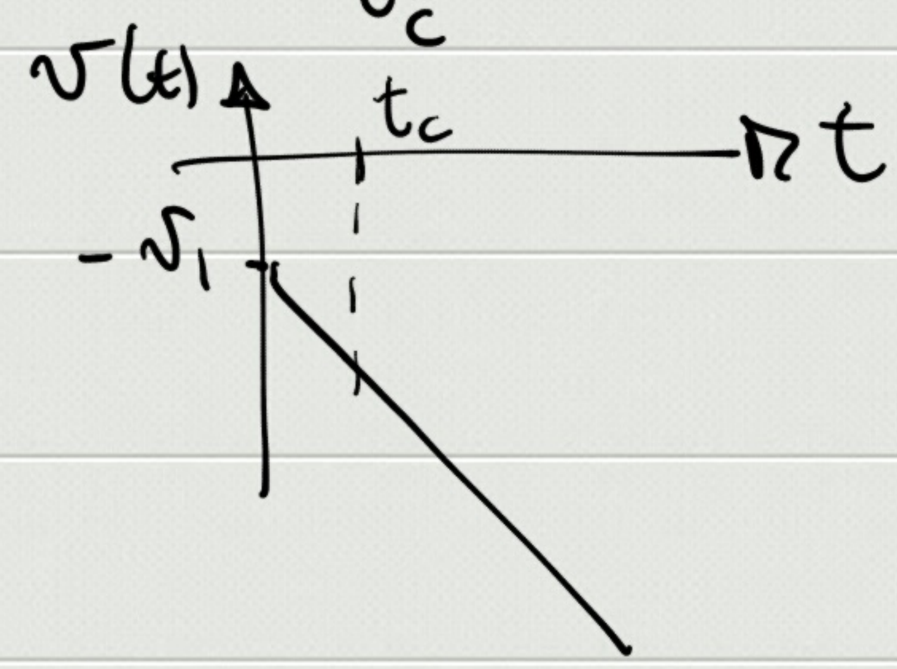
$x_0 = h \qquad v_0 = -v_1 < 0 \qquad (v_1 > 0)$

$$x(t) = h - v_1 t - \frac{1}{2} g t^2$$

$$v(t) = -v_1 - g t$$

$$v^2(x) = v_1^2 - 2g(x-h)$$





$$g t^2 + 2 v_1 t + 2(x-h) = 0$$

$$t = \frac{-v_1 \pm \sqrt{v_1^2 - 2g(x-h)}}{g}$$

$$t_c = t(x=0) = \frac{-v_1 \cancel{\pm} \sqrt{v_1^2 + 2gh}}{g}$$

$$v_c = v(t_c) = -v_1 - g t_c$$

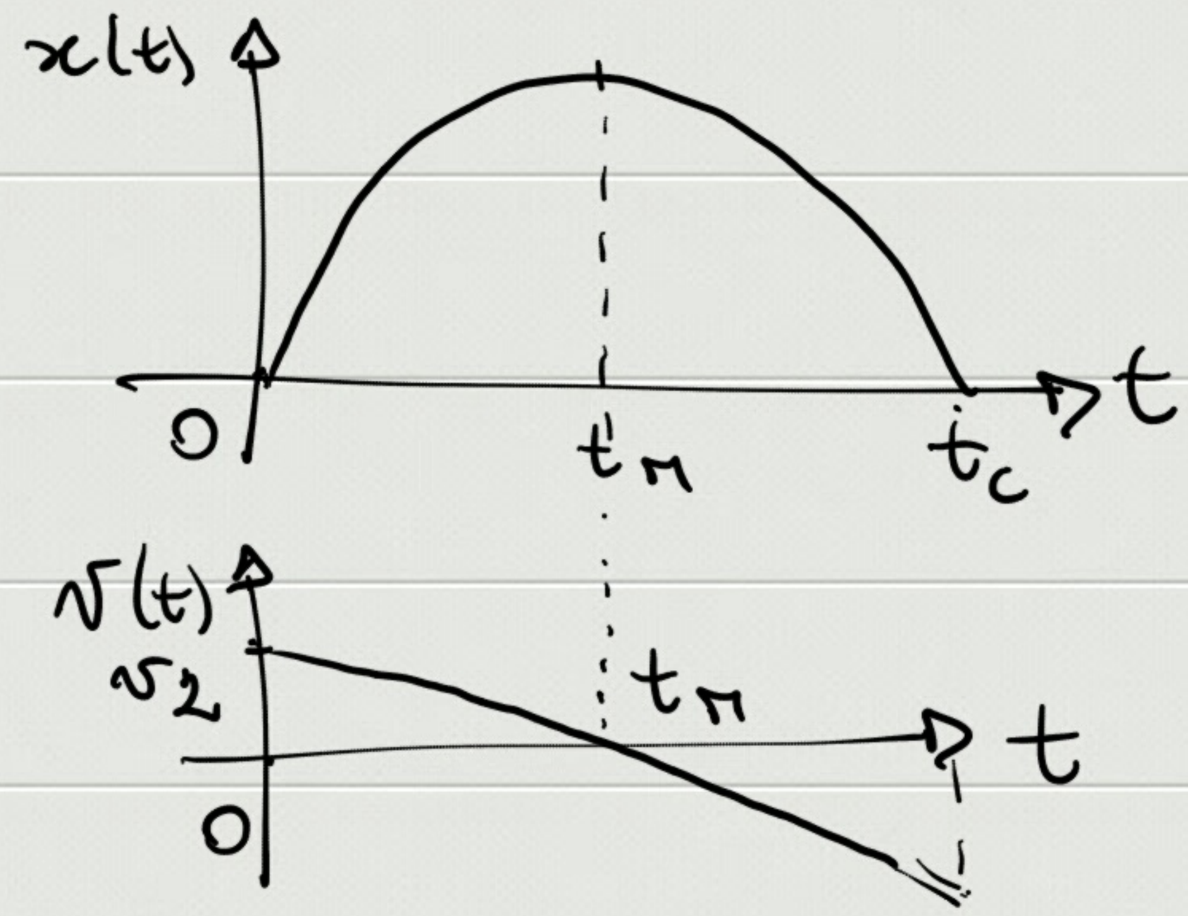
$$v_c = v(x=0) = -\sqrt{v_1^2 + 2gh}$$

$x_0 = 0 \qquad v_0 = v_2 > 0$

$$x(t) = v_2 t - \frac{1}{2} g t^2$$

$$v(t) = v_2 - g t$$

$$v^2(x) = v_2^2 - 2gx$$



$$t_M : \; v(t_M) = 0 \Rightarrow v_2 - g t_M = 0 \Rightarrow \boxed{t_M = \frac{v_2}{g}}$$

$$t_c = 2 t_M$$

$$g t^2 - 2 v_2 t - 2x = 0$$

$$\Rightarrow t = \frac{v_2 \pm \sqrt{v_2^2 + 2gx}}{g}$$

$$v_c = v(2 t_M) = v\left(\frac{2 v_2}{g}\right) = v_2 - g \frac{2 v_2}{g} = - v_2$$